INCH-POUND

MIL-S-20708/2E
13 March 1989

SUPERSEDING
MIL-S-20708/2D
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO, CONTROL TRANSMITTER, TYPE 11CX4E

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708.

TABLE I. Requirements.

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	115	volts	±1%
Primary Current	31.0	milliamps	maximum
Primary Power	0.59	watts	maximum
Impedance:			
Zro	3720-4700	ohms	min.-max.
Zss	410-500	ohms	min.-max.
Impedance Angle:			
Zro	80.0-84.0	degrees	min.-max.
Zss	17.0-26.0	degrees	min.-max.
Transformation Ratio	0.783	------	±2%
Phase Shift (Lead)	4.5	degrees	±1.5
Electrical Error	7.0	minutes	maximum
Null Voltage:			
Total	75.0	millivolts	maximum
Fundamental	45.0	millivolts	maximum
Friction Torque	0.07	ounce-inches	maximum
Radial Play	0.0006	inches	maximum
End Play	0.0010	inches	±0.0005
Temperature Rise	20.0	degrees (C)	maximum

Ⓔ denotes changes

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TABLE II. Military part number variant characteristics.

Military Part No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U +.0000 −.0002 Ⓔ
M20708/2-01E	.555	.500	1.732	Splined	Terminal Board	.1872
M20708/2-02E	.368	.312	1.630	Plain & Slotted	Wire Leads	.1200
M20708/2-03E	.430	.375	↓	↓	↓	↓
M20708/2-04E	.493	.437				
M20708/2-05E	.555	.500				
M20708/2-06E	.618	.562				

1/Military part number M20708/2-01E shall be the NATO standard.

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Military part number cross-references.

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBERS
MIL-S-20708/2C	MIL-S-20708/2B	
M20708/2C-001	M20708/2-001	M20708/2-01E
M20708/2C-002		M20708/2-02E
M20708/2C-003		M20708/2-03E
M20708/2C-004		M20708/2-04E
M20708/2C-005		M20708/2-05E
M20708/2C-006		M20708/2-06E

NOTE: All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification. Part number changes do not affect form, fit, or function of the synchros listed therein.

Custodians:
Army-AR
Navy-AS
Air Force-85

Review Activities:
Army-MI,AV
Navy-OS
Air Force-11,17,99
DLA-ES

User Activities:
Navy-MC,CG

Preparing Activity:
Navy-AS

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